Celloras

Make cell annotation a conversation.

An AI research agent for plant single-cell data.



7,496 nuclei. One research conversation.

The data is there.
Researchers still have to connect it.

Gene activity

Glyma.01G224900
Glyma.17G030700

Marker tables

Which genes
support a label?

Papers

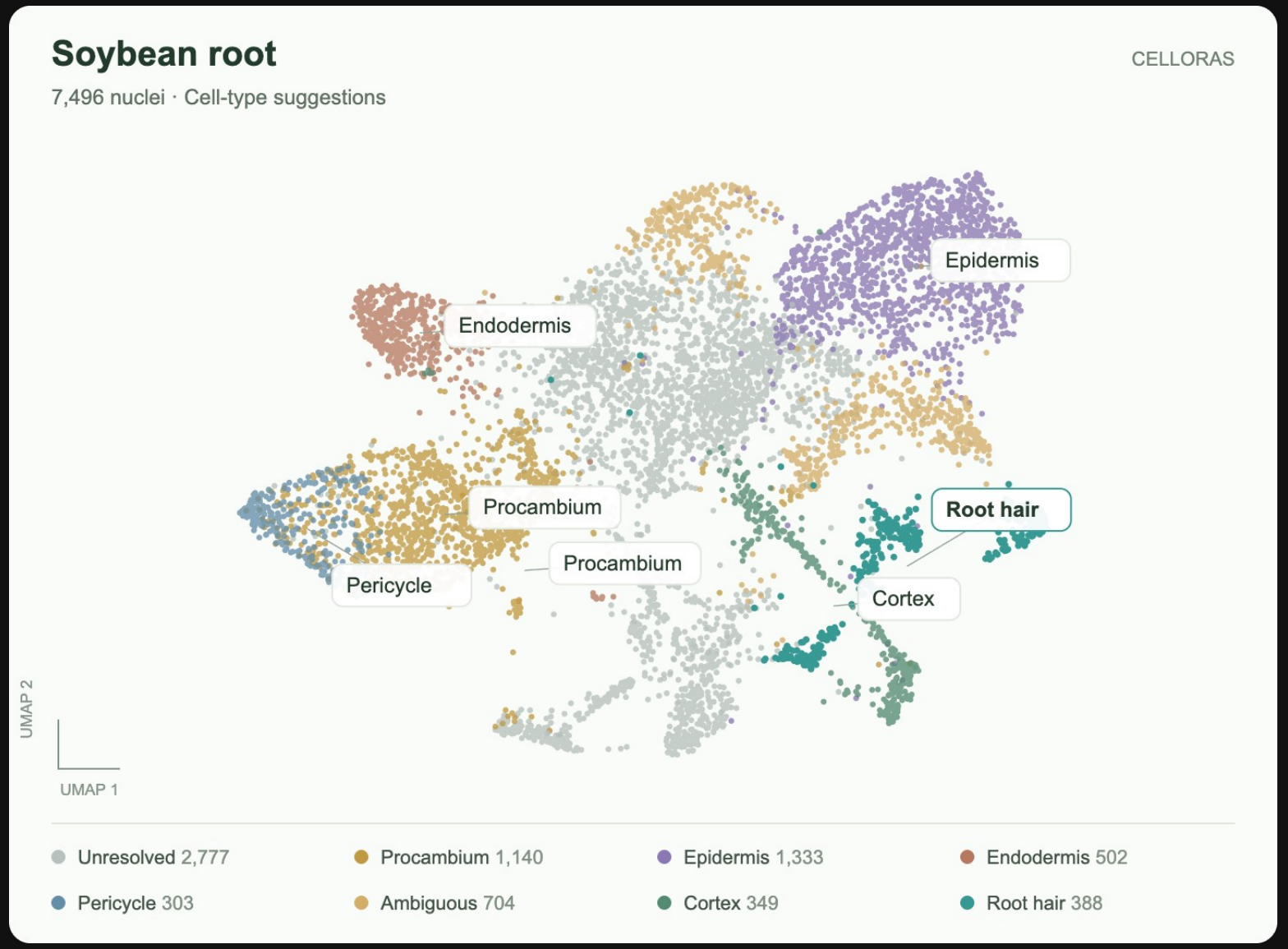
The right plant?
The right tissue?

Past decisions

What did we decide,
and what changed?

One label depends on all of them.

"Annotate all groups and show the evidence."



One request. Twelve groups.

7 suggested

1 ambiguous

4 unresolved

Why Root hair?

388 nuclei

Two supporting marker genes

Glyma.01G224900
Glyma.17G030700

Inspect the measurements,
then open the source.



Dot size: fraction expressing · color: mean expression

Every suggestion comes with a source.

Zetaris

Published marker panel
↓
Versioned reference query
↓
Saved analysis receipt

42 associations, checked against the reference used in the analysis.



Research changes.
Memory should keep up.

Computed candidate

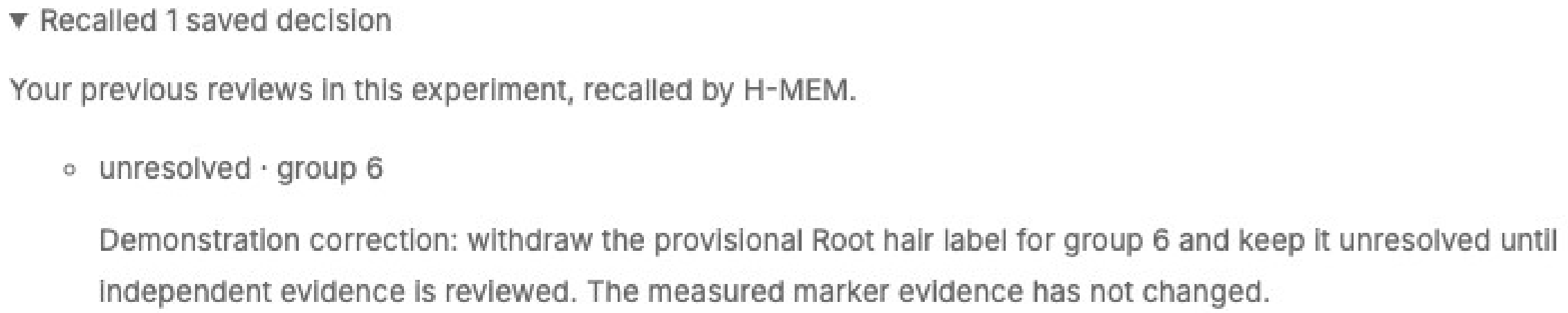
Root hair

→

Researcher's latest decision

Unresolved

Meterless H-MEM recalls the decision, its reason and its source.



The agent does the work.
The researcher makes the call.



Next.js + Keycloak · Project-owned data · PostgreSQL

Celloras

Less reconstruction. More research.

Working today

Soybean annotation, evidence and decision recall.

Next milestone

Independent validation, then researchers’ own plant data.

Try the soybean workflow

celloras.com

Validation & current scope

2,952

numerical comparisons passed

Independent base-R implementation compared with the original count data.

Live workflow passed

Annotation → evidence → review → recall → corrected recall

Biological validation is next.

The current panel comes from the same study. These tests establish computation and workflow behavior.

Sources & implementation

Soybean data	Zhang et al. · GSE270392
Scientific method	Reference panel, scoring and limitations
Sponsor integrations	Zetaris + official Meterless H-MEM Engine
Code & demo	Public application branch and walkthrough